

Original research

Calibration and refinement of ACMG/AMP criteria for variant classification with BayesQuantify

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ABSTRACT

Background Improving the precision and accuracy of variant classification in clinical genetic testing requires further specification and stratification of the American College of Medical Genetics/Association of Molecular Pathology (ACMG/AMP) criteria. While the ClinGen Bayesian framework enables quantitative evidence calibration for selected criteria, standardised tools to optimise evidence thresholds and refine ACMG/AMP criteria remain underdeveloped.

Methods To address this need, we developed *BayesQuantify*, an R package that provides a unified tool for quantifying evidence strength for the ACMG/AMP criteria based on the Bayesian framework. *BayesQuantify* accepts a variant classification file as input and automatically calculates the odds of pathogenicity for each evidence strength, incorporating a user-provided prior probability of pathogenicity. Through bootstrapping, *BayesQuantify* generates thresholds by aligning the 95% lower bound of positive likelihood ratio/local positive likelihood ratio with the odds of pathogenicity for different evidence strengths. Three independent datasets derived from ClinVar, HGMD and gnomAD were used to evaluate the utility of *BayesQuantify*.

Results *BayesQuantify* supports the calibration of both categorical and continuous ACMG/AMP evidence. Specifically, we replicated the PP3/BP4 thresholds for four computational tools recommended by ClinGen. Our analysis also indicated that the PM2 criterion can reach 'supporting,' or 'moderate,' evidence, varying by prior probability. Importantly, we established thresholds for supporting, moderate and strong evidence for in-silico tools, thereby expanding the application of PP3/BP4 criteria for missense variants in the *PTEN* gene.

Conclusion *BayesQuantify* is a user-friendly tool that enhances the flexibility and reproducibility of ACMG/AMP criteria refinement, thus improving the accuracy and consistency of variant classification. The package is freely available at <https://github.com/liusihan/BayesQuantify>.

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ In 2018, the ClinGen Sequence Variant Interpretation Working Group proposed a Bayesian framework that presents opportunities for refining the American College of Medical Genetics/Association of Molecular Pathology (ACMG/AMP) criteria.
- ⇒ Further, researchers have successfully quantified and extended the strengths of evidence for PS1/PM5, PS3/BS3, PP3/BP4 and PS4 criteria according to this framework.
- ⇒ However, to our knowledge, there is a lack of freely available and standardised tools that provide similar functionalities for the academic community.

WHAT THIS STUDY ADDS

- ⇒ The *BayesQuantify* enables objective, consistent refinements for both categorical and continuous ACMG/AMP evidence by aligning the positive likelihood ratio/local positive likelihood ratio values with the odds of pathogenicity for different evidence levels, thus providing a unified resource for quantifying evidence strength for ACMG/AMP criteria and classifying variants based on the Bayesian framework.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ The *BayesQuantify* will enable the shift from a predominantly subjective assessment to a more quantitative and objective approach.
- ⇒ By calibrating the ACMG/AMP criteria, *BayesQuantify* can facilitate the reclassification of variants of uncertain significance and improve diagnostic accuracy.



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INTRODUCTION

Interpreting sequence variants in human genes is a pivotal aspect of diagnosing and managing genetic diseases.^{1,2} However, this task presents considerable challenges due to the complexity and diversity of the genomic data and the lack of standardisation and consistency across laboratories.³ To address these issues, the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) have developed a set of guidelines for variant interpretation, comprising 28 criteria which support or refute the pathogenicity or

benignity of a variant.⁴ These criteria are grouped into five strength levels of evidence (stand-alone, very strong, strong, moderate and supporting) and are combined according to specific rules to assign a variant into one of five categories: pathogenic (P), likely pathogenic (LP), uncertain significance (US), likely benign (LB) and benign (B). Since their introduction, the ACMG/AMP guidelines have facilitated the genetic diagnosis of suspected inherited disorders and have been widely adopted by clinical laboratories and researchers as the standard for variant interpretation and reporting.^{5–8}

In 2018, the ClinGen Sequence Variant Interpretation Working Group (ClinGen SVI) proposed a Bayesian framework that transformed the ACMG/

AMP guidelines into a probabilistic system.⁹ This framework assumes four levels of evidence and exponentially scaled odds of pathogenicity (OP). It assigns a posterior probability of pathogenicity (Post_P) to a variant by multiplying the OP for each level of evidence ($O_{PVSt}, O_{PSst}, O_{PM}, O_{PSu}, O_{BVSt}, O_{BSt}, O_{BM}, O_{BSu}$) with a prior probability of pathogenicity (Prior_P). The resulting posterior probability is then compared with probability thresholds to categorise variants: (1) the pathogenic combinations would have $Post_P > 0.99$, (2) the Post_P of the likely pathogenic combinations would be $0.90 < Post_P \leq 0.99$, (3) the Post_P of the likely benign combinations would be $0.001 \leq Post_P < 0.10$, and (4) the one benign category would have $Post_P < 0.001$. By modelling the ACMG/AMP guidelines with a Prior_P of 0.10 and O_{PVSt} set to 350, 15 out of 17 existing ACMG/AMP evidence combinations (except BA1) were found to be mathematically consistent with the overall framework, validating the approach adopted by the ACMG/AMP.

The Bayesian framework presents opportunities for refining evidence and combining rules, as well as supporting efforts to automate components of variant pathogenicity assessments. Researchers have quantified and extended the strengths of evidence criteria such as PS1/PM5, PS3/BS3 and PP3/BP4 according to the Bayesian framework.^{10–16} Importantly, thresholds for PS4, PS4_Moderate and PS4_Supporting have been quantified by aligning positive likelihood ratio (LR^+) and local positive likelihood ratio (lr^+) values with theoretical OP for each evidence strength in our prior study.¹⁷ Our study shows that the refinement of PS4 improves the identification of pathogenic variants, supporting that the specification and stratification of evidence is essential to establish an objective, unbiased and consistent variant classification.^{7,8,18} However, there is a lack of existing software and tools tailored to quantify the strength of evidence and establish appropriate thresholds to refine the ACMG/AMP criteria.

In this study, we develop *BayesQuantify*, an R package that undergoes the ClinGen Bayesian framework for quantifying the strength of evidence for ACMG/AMP criteria. *BayesQuantify* provides functions to estimate OP values for each level of evidence strength and the posterior probability of pathogenicity for the ACMG/AMP combining rules. Furthermore, *BayesQuantify* facilitates the calculation of LR^+ and lr^+ for various tested cut-offs, aligning the 95% lower boundary of LR^+/lr^+ with OP values to establish thresholds. Additionally, users can employ *BayesQuantify* to classify variants in accordance with the ACMG/AMP combination rules and Bayesian framework, as well as to visualise both the data and statistical results.^{4,9} The *BayesQuantify* is available from GitHub at <https://github.com/liusihan/BayesQuantify>.

METHODS

BayesQuantify workflow overview

The functions available through *BayesQuantify* were designed to quantify the evidence strength and thresholds following the algorithm proposed by ClinGen SVI. The package requires one essential input: a *data.frame* containing variants classification results, including the variants, ACMG/AMP evidence, variant classification and tested features (e.g., online supplemental tables S1 and S2). The step-by-step usage of *BayesQuantify* consists of five main steps as follows (figure 1):

1. Step 1: Calculation of OP and Post_P for each evidence strength: The SVI Working Group has provided an Excel spreadsheet to calculate Post_P from Prior_P and OP from pathogenic and benign evidence categories.⁹ Based on this spreadsheet, users

can manually adjust the values of Prior_P and O_{PVSt} to obtain the OP values corresponding to each level of evidence strength. However, this process was computationally inefficient and prone to user bias. To overcome these limitations, we developed *auto_select_postp()*, a core algorithm in *BayesQuantify*, which systematically evaluates O_{PVSt} values over a plausible range (1–10 000) in combination with user-defined Prior_P values. For each iteration, it calculates Post_P using Bayes' rule and validates compliance with the ACMG/AMP rules for combining evidence and predefined pathogenicity probability thresholds (Equation 1 and 2). The output identifies the minimum O_{PVSt} value that satisfies these constraints, ensuring robust, reproducible OP and Post_P calculation.

$$OP = O_{PVSt} \left(\frac{NPVSt}{1} + \frac{NPSst}{2} + \frac{NPM}{4} + \frac{NPSu}{8} - \frac{NBSu}{8} - \frac{NBM}{4} - \frac{NBSt}{2} - \frac{NBVSt}{1} \right) \quad (1)$$

$$Post_P = \frac{OP \times Prior_P}{((OP-1) \times Prior_P + 1)} \quad (2)$$

2. Step 2: Variants of US (VUS) subclassification: The majority of human genomic variants are rare, resulting in a high prevalence of VUS in clinical classifications. VUSs encompass a broad spectrum of pathogenicity probability, ranging from 10% to 90%. To address this variability, we developed the *VUS_classify()* function, which categorises VUS into six confidence tiers (hot, warm, tepid, cool, cold and ice cold) according to the Association for Clinical Genomic Science Best Practice Guidelines (<https://www.acgs.uk.com/quality/best-practice-guidelines/#VariantGuidelines>). The hot, warm and tepid tiers encompass VUS for which additional testing could alter their classification and that are likely to be disease-causing. To ensure a robust and reliable truth set for evidence calibration, we recommend excluding these variants from the dataset to prevent confounding with potentially pathogenic variants. This stratification approach guarantees a high-confidence dataset for calibrating evidence thresholds, while preserving both biological and clinical relevance.

3. Step 3: Calculating LR^+/lr^+ : Within the ClinGen Bayesian framework, the OP quantifies the probability of observing specific evidence if a variant is pathogenic. To quantify evidence thresholds, *BayesQuantify* provided functions to calculate the LR^+/lr^+ using the following methodologies:

For binary variables (e.g., variant marked with PM2 or not): The *LR()* function computes various performance metrics by evaluating the presence or absence of specific evidence among disease-causing and non-pathogenic variants. These metrics include true positive (TP), false positive (FP), true negative (TN), false negative (FN), overall accuracy, true positive rate (sensitivity), true negative rate (specificity), positive predictive value (PPV), negative predictive value (NPV), F1 score and LR^+ (Equation 3). The 95% CI for LR^+ is estimated using bootstrapping in the R package *bootLR*.¹⁹ The lower bound of LR^+ (LR^+_{LB}) is then used to determine evidence strength by comparing it with the OP thresholds generated in Step 1.

$$LR^+ = \frac{TP/(TP+FN)}{1-TN/(TN+FP)} \quad (3)$$

For quantitative variables (e.g., scores from in-silico tools used for tagging PP3/BP4): the *local_bootstrapped_lr()* function first sorts all unique tested cut-offs and positions each cut-off at the centre of a user-defined sliding window (default: 0.01). The local posterior probability of pathogenicity and benignity is then calculated for each tested cut-off within the interval, considering a minimum quota of disease-causing and non-pathogenic variants (Equation 4). Additionally, the one-sided 95% confidence bound for each estimated lr^+ is determined through bootstrapping iterations. The *get_lr_threshold()* function can determine

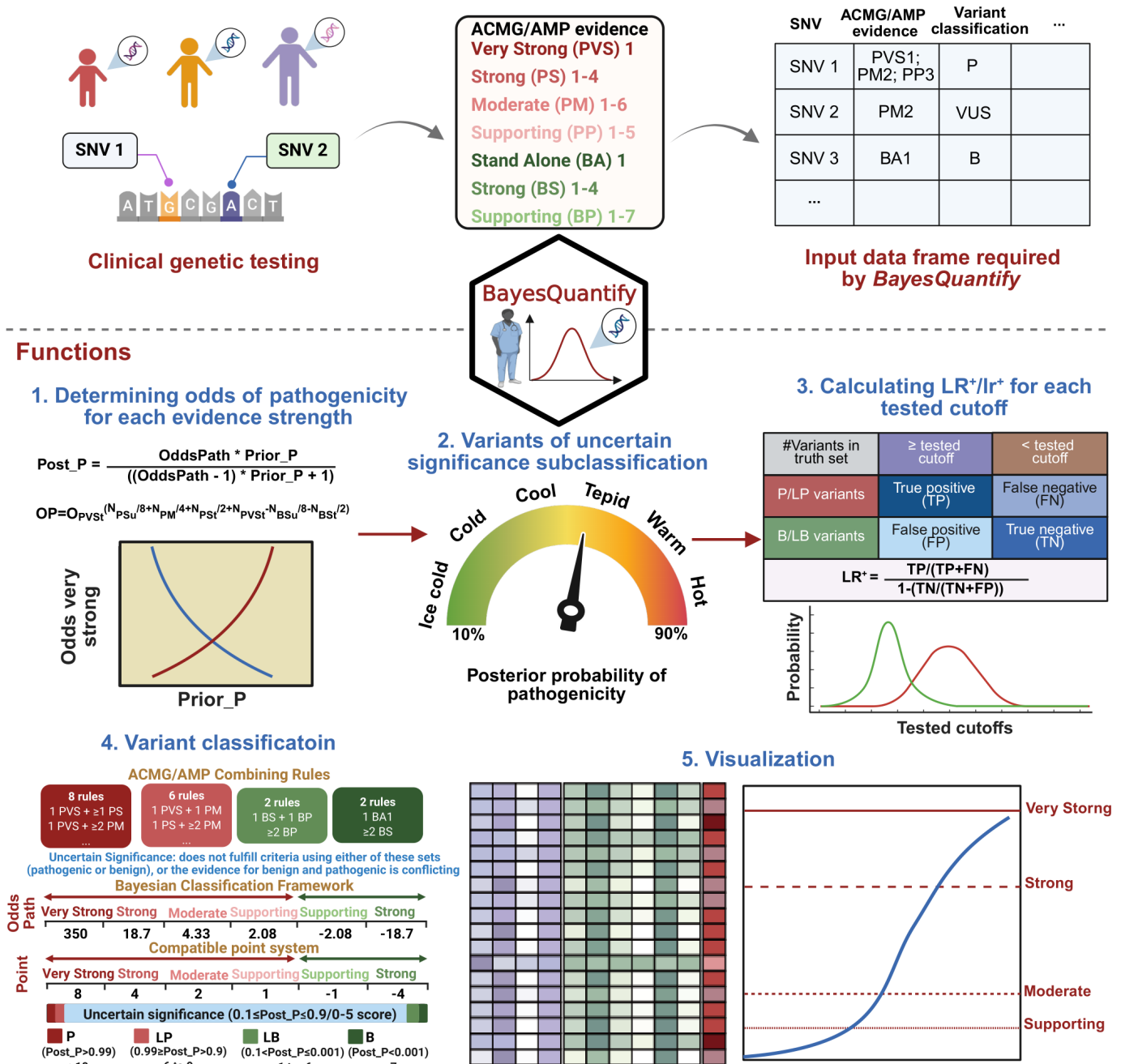


Figure 1 : Schematic overview of BayesQuantify. BayesQuantify begins with the input of variant classification results dataset. Subsequently, it computes the positive likelihood ratio (LR⁺) or local positive likelihood ratio (lr⁺), which are then aligned with predefined odds of pathogenicity (OP) for each evidence strength to refine evidence. ACMG/AMP, American College of Medical Genetics/Association of Molecular Pathology; FN, false negative; FP, false positive; TN, true negative; TP, true positive; VUS, variants of uncertain significance; Prior_P: prior probability of pathogenicity; Post_P: posterior probability of pathogenicity.

the thresholds for each evidence level by mapping the lower bound of local posterior probability with the thresholds generated in Step 1.

$$\text{Local posterior probability} = \frac{\#P/LP \text{ variants in interval}}{\#P/LP \text{ variants in interval} + (\text{Weight} \times \#non - \text{pathogenic variants in interval})} \quad (4)$$

4. Step 4: Variant classification: BayesQuantify offers two functions for variant classification: The *ACMG_Classification()* function categorises variants into P, LP, B, LB or VUS using predefined evidence combinations from the ACMG/AMP guidelines. In parallel, the *BCF()* function applies to a Bayesian

classification framework to classify variants by calculating posterior probability of pathogenicity with user-provided Prior_P and OP for each evidence level.

5. Step 5: Visualisation: To improve user-friendliness, BayesQuantify includes several visualisation functions: (1) the *multi_plot()* function generates composite plots to display variant characteristics (e.g., allele frequency (AF), variant consequence) in the input data.frame; (2) the *heatmap_LR()* function creates heatmaps to visualise the distribution of positive likelihood ratios across tested cut-offs, highlighting thresholds where LR⁺ values meet or exceed OP bounds for pathogenicity or

benignity; (3) the *plot_lr()* function plots local posterior probability curves for quantitative variables, showing the distribution of lr^+ ; and (4) the *evidence_corplot()* function generates visualisations of evidence correlations.

All these steps are implemented within the R programming environment to consolidate all intermediate data generated throughout the workflow, ensuring transparency and user-friendly operation. For a detailed explanation of each step with code and examples, see the website of the package (<https://github.com/liusihan/BayesQuantify>).

Data sets

The ClinVar 2019 dataset

To evaluate the performance of *BayesQuantify*, the ClinVar 2019 dataset, compiled by Pejaver *et al*, was obtained from <https://zenodo.org/records/8347415>.¹⁰ This dataset includes missense variants with an AF below 0.01 in the Genome Aggregation Database (gnomAD V2.1) and from genes with at least one pathogenic variant of any type in ClinVar.²⁰ In total, 11 834 variants (5705 P/LP and 6129 B/LB variants) from 1913 genes are included in this dataset (online supplemental table S1).

The ClinGen Curated Variants dataset

The ClinGen Curated Variants dataset was retrieved from the ClinGen Evidence Repository as of 7 April 2024 (<https://repo.clinicalgenome.org/evrepo/>). This dataset contains

expert-curated assertions on variant pathogenicity, along with supporting evidence summaries.¹⁸ It includes classification summaries for 6768 variants across 73 diseases, with 1850 P, 1463 LP, 679 LB, 775 B and 2001 US variants (figure 2 and online supplemental table S2). Variants tagged with BA1 were excluded, resulting in the removal of 664 variants from downstream analysis.

The PTEN dataset

To perform the calibration for the *PTEN* gene (MIM# 601728), we constructed a dataset, referred to as the PTEN dataset, which included variants classified as disease-causing and non-pathogenic. Variants were sourced from three databases: ClinVar (V.20241120),²¹ HGMD (2024q1)²² and gnomAD (V.4.1).²⁰ These variants were filtered out based on the following criteria: (1) Only missense variants located in the *PTEN* (NM_000314.8) gene with an AF below 0.01 in gnomAD were retained; (2) for variants observed exclusively in gnomAD, only those variants annotated as ‘PASS’ in the ‘FILTER’ column were included; (3) variants with conflicting pathogenicity annotations between ClinVar (P/LP) and HGMD (DM/DM?) were excluded (except variants reviewed by expert panel); and (4) variants classified as uncertain significance or with conflicting classifications in ClinVar were also removed. Disease-causing variants were defined as P/LP variants, either with a review status of ‘criteria_provided, multiple_submitters, no_conflicts,’ or

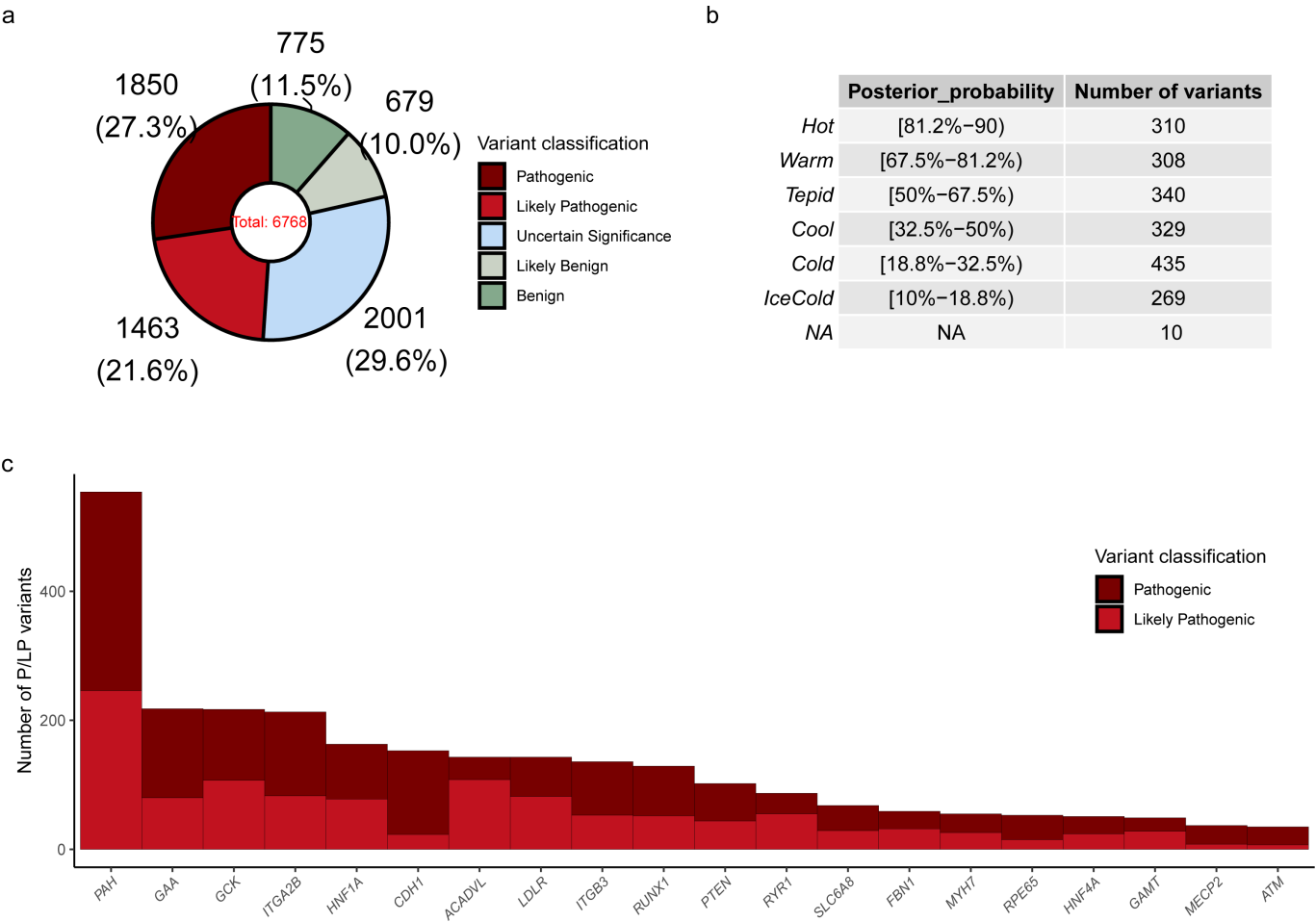


Figure 2 : Characteristics of the 6,768 variants in the ClinGen Curated Variants dataset. (a) Summary of classification for the 6,768 variants. (b) Further subclassification of variants of uncertain significance (VUS) to hot, warm, tepid, cool, cold, and ice-cold groups, according to the ACGS Best Practice Guidelines. (c) Distribution of pathogenic (P) and likely pathogenic (LP) variants across the top twenty genes.

'reviewed_by_expert_panel,' or those with consistent classifications between ClinVar and HGMD. Non-pathogenic variants were included as variants classified as B/LB variants in ClinVar or rare variants only observed in gnomAD (online supplemental table S3).

Estimation of the prevalence of pathogenic variants

To our knowledge, three approaches have been described in previous studies to estimate the prior probability of pathogenicity: (1) Tavtigian *et al* assumed a prior of 10% based on the context of identifying a pathogenic variant among candidate variants in clinical genetic testing (commonly used)^{11–13}; (2) Pejaver *et al* estimated a lower prior (4.41%) using ClinVar and gnomAD data; and (3) another approach involves calculating the prior probability using cohort data. For example, Bhat *et al*²³ established gene-specific and phenotype-specific priors (0.1–23.2%) using UK Biobank data, and in our previous work, we calculated priors using the Chinese Deafness Genetics Consortium cohort data (9.31–28.5%).¹⁷

The prior probability for the ClinVar 2019 dataset was derived from the work of Pejaver *et al*. For the ClinGen Curated Variants dataset and the PTEN dataset, dataset-specific prior probability was calculated following a similar approach to that of Pejaver *et al*. To estimate the prevalence of pathogenic variants in the ClinGen Curated Variants dataset, we performed the following steps to generate a reference dataset from gnomAD: (1) 332 531 variants were extracted from gnomAD (V.4.1), focusing on coding regions of genes with at least one P/LP variant in the ClinGen Curated Variants dataset; (2) variants with an AF <0.01 and those annotated as 'PASS' in the 'FILTER' column were retained; and (3) P/LP variants present in ClinVar (V.20241120) or HGMD (2024q1) were removed. The final reference dataset consisted of 133 270 potentially non-pathogenic variants. Based on this, we estimated the prevalence of pathogenic variants in the ClinGen Curated Variants dataset to be 2.49% (3313/133 270).

For estimating the prevalence of pathogenic variants in the PTEN dataset, we used a combined dataset from ClinVar (V.20241120), HGMD (2024q1) and gnomAD (V.4.1). Variants in the *PTEN* gene with an AF >0.01 in gnomAD, as well as those with conflicting pathogenicity annotations between ClinVar (P/LP) and HGMD (DM/DM?), were excluded. The prevalence of pathogenic variants in this dataset was found to be 4.68% (1319/28 171).

We caution that estimating pathogenic variant prevalence using gnomAD as a reference set may underestimate true values due to potential residual contamination by undocumented P/LP variant. In the context of a substantially lower prior probability, a higher O_{PVSt} threshold is required to attain the equivalent posterior probability of pathogenicity.

Calibration of computational tools in the PTEN dataset

To systematically evaluate and calibrate computational tools, we annotated variants using precomputed scores from the Database of Human Nonsynonymous SNPs (dbNSFP 4.7a) via the variant effect predictor.^{24 25} A total of 38 prediction scores from the dbNSFP database, covering the entire human genome, were compared. All genomic positions for these precomputed scores were based on the GRCh38/hg38 reference genome. To standardise the assessment, we employed a rank score methodology, which scaled the raw prediction scores to a 0–1 range, reflecting the proportion of variants deemed less damaging. The

performance of these 38 tools was then evaluated in the PTEN dataset using receiver-operating characteristic (ROC) curves, generated with the R package pROC.²⁶

RESULTS

Generating OP values and high-confidence datasets using *BayesQuantify*

Comprehensive OP values for ACMG/AMP criteria were established across three distinct datasets through the *auto_select_postp()* function in *BayesQuantify*. For the ClinVar 2019 dataset (Prior_P=4.41%, estimated by Pejaver *et al*), pathogenic evidence required OP values ≥ 2.41 (supporting), ≥ 5.80 (moderate), ≥ 33.65 (strong) and ≥ 1132 (very strong), while benign evidence thresholds were ≤ 0.415 , ≤ 0.172 , ≤ 0.03 and ≤ 0.001 , respectively. The ClinGen Curated Variants dataset (Prior_p=2.49%) exhibited elevated pathogenic thresholds ($O_{PSu} \geq 2.66$, $O_{PM} \geq 7.06$, $O_{PSt} \geq 49.9$, $O_{PVSt} \geq 2490$) with corresponding benign thresholds at $O_{BSu} \leq 0.376$, $O_{BM} \leq 0.142$, $O_{BSt} \leq 0.02$ and $O_{BVSt} \leq 0.0004$. For the PTEN dataset, the OP values were: $O_{PSu} \geq 2.384$, $O_{PM} \geq 5.682$, $O_{PSt} \geq 32.28$, $O_{PVSt} \geq 1042$, $O_{BSu} \leq 0.42$, $O_{BM} \leq 0.176$, $O_{BSt} \leq 0.031$ and $O_{BVSt} \leq 0.001$, with a Prior_P of 4.68%.

To accurately refine the ACMG/AMP criteria, the *VUS_classify()* function was used to categorise VUS into six levels, generating high-confidence datasets by including cool, cold and ice cold VUS. In the ClinGen Curated Variants dataset, this approach resulted in 5137 variants, comprising 1847 P, 1463 LP, 123 B, 673 LB and 1031 VUS variants. Correspondingly, the ClinVar 2019 dataset encompassed 5705 disease-causing variants and 6129 non-pathogenic variants, while the PTEN dataset consisted of 326 disease-causing variants and 69 non-pathogenic variants.

Validating the utility of *BayesQuantify* for evidence refinement and variant classification

To assess the accuracy of *BayesQuantify* in refining evidence for variant classification, we replicated the calibration of four recommended computational tools (BayesDel, MutPred2, REVEL and VEST4) using the ClinVar 2019 dataset, as proposed by Pejaver *et al* for determining PP3 and BP4 evidence levels. Using 10 000 bootstrap iterations, we calculated local posterior probability for each unique score generated by these tools. Our analysis confirmed that all four tools achieved thresholds for strong evidence of pathogenicity (BayesDel ≥ 0.53 , MutPred2 ≥ 0.925 , REVEL ≥ 0.921 , VEST4 ≥ 0.96). For benignity, BayesDel (≤ -0.36) and VEST4 (≤ 0.32) reached moderate evidence thresholds, while MutPred2 (≤ 0.023) and REVEL (≤ 0.047) met strong evidence thresholds (figure 3, online supplemental figure S1 and table S4). These results align with Pejaver *et al*'s findings, validating the reliability of *BayesQuantify* in evidence calibration. Additionally, local posterior probability was estimated using a prior probability of pathogenicity of 10%. The related evidence thresholds are presented in online supplemental table S5, and local posterior probability curves are presented in online supplemental figure S2.

We further evaluated *BayesQuantify*'s ability to classify variants into five categories (P, LP, B, LB and VUS) according to the ACMG/AMP guidelines and the Bayesian classification framework. Using the ClinGen Curated Variants dataset, *BayesQuantify* demonstrated 99.02% concordance (n=6702 variants) with existing classifications (online supplemental table S6). Of the 66 discordant variants, discrepancies were due to expert curation that extended beyond guideline-based evidence. A representative example is the *CDH23* variant NM_022124.6:c.380A>G

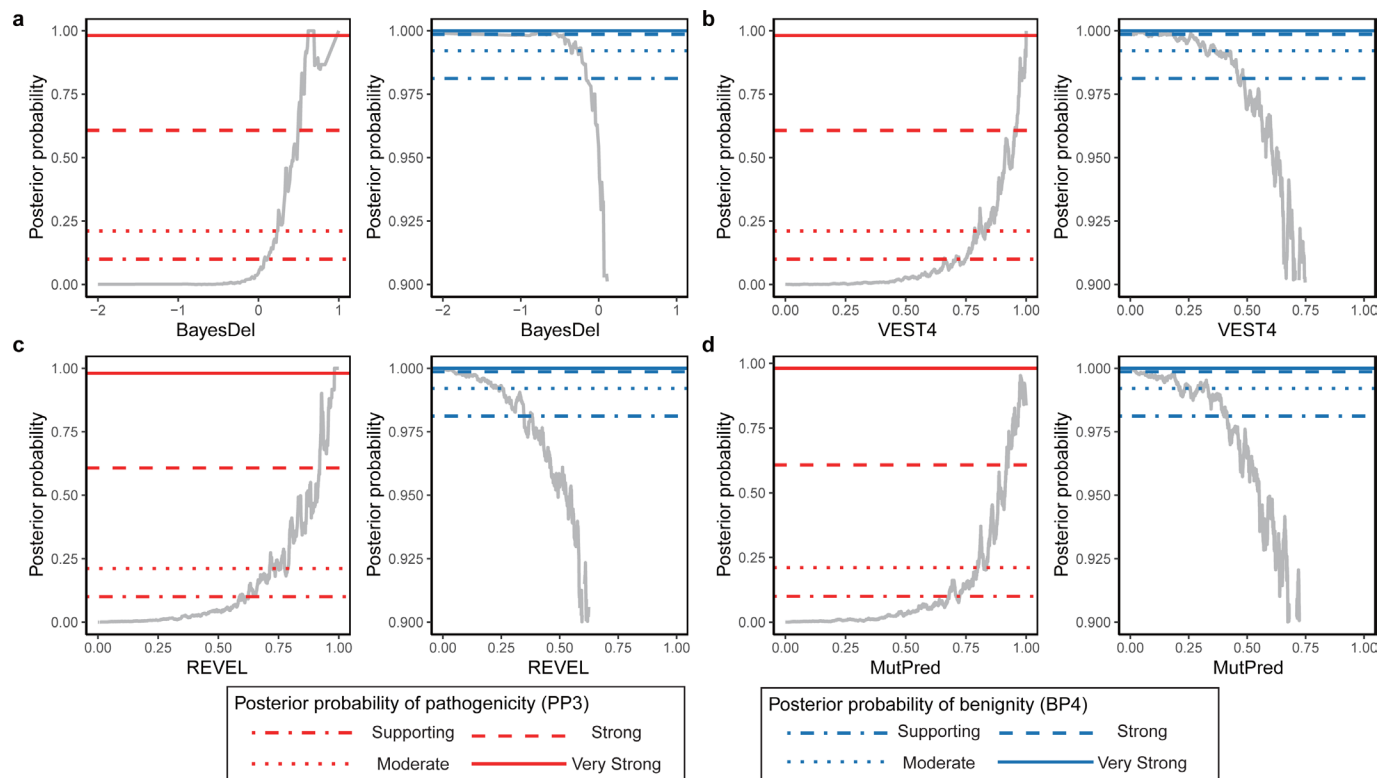


Figure 3 : Local posterior probability curves of pathogenicity and benignity for various BayesDel, VEST4, REVEL and MutPred2 scores in the ClinVar 2019 dataset. The horizontal lines depict the posterior probability thresholds for supporting, moderate, strong, and very strong evidence (red for pathogenicity and blue for benignity). The grey curve represents the lower boundary of the one-sided 95% confidence interval for each score, calculated using 10 000 bootstrapping from the ClinVar 2019 dataset. The points where the grey curves intersect the horizontal lines represent the thresholds for the corresponding intervals. REVEL, rare exome variant ensemble learner.

(p.Asp127Gly), for which the ClinGen Hearing Loss Expert Panel provided the following evidence: PM2 Supporting, PP3, PM3 and PP4. According to the ACMG/AMP combination rules, this combination supports a classification of VUS, which aligns with the output generated by *BayesQuantify*. Notably, based on expert judgement, the ClinGen Hearing Loss VCEP ultimately classified this variant as likely pathogenic for autosomal recessive Usher syndrome. This serves as a reminder that, as emphasised by the ACMG/AMP guidelines, final variant classification requires manual review and expert curation. To improve clinical consistency and accuracy in variant classification, *BayesQuantify* supports the integration of expert-curated variants into the classification process.

Evaluating the clinical weight of PM2 criteria in variant pathogenicity classification

The 2015 ACMG/AMP guidelines originally defined PM2 as a moderate pathogenic criterion, applicable when a variant is absent or extremely rare (e.g., recessive conditions) in population databases such as gnomAD. However, concerns arose about overestimating its evidence weight, as large-scale studies (e.g., ExAC) revealed that 54% of rare variants occurred only once, suggesting that rarity alone is insufficient to infer pathogenicity. To address this concern, we evaluated the clinical weight of PM2 criteria using *BayesQuantify*. In the ClinGen Curated Variants dataset, PM2 was designated as moderate evidence and supporting evidence in accordance with gene-/disease-specific guidelines and emerged as the most frequently applied pathogenic criterion, being utilised in 3072 P/LP variants, 580 VUS

and 114 B/LB variants, respectively (online supplemental figure S3). We evaluated whether PM2 and PM2_Supporting met the relative OP threshold for moderate (7.064:1) and supporting (2.658:1) evidence of pathogenicity. A total of 3766 variants were tagged with PM2_Supporting, including 3072 P/LP variants and 694 non-pathogenic variants. The accuracy was 0.819, PPV was 0.816, sensitivity was 0.928, specificity was 0.620, and LR^+ was 1.497 (95% CI 1.443 to 1.555), which was lower than the O_{PSu} threshold for the supporting evidence level (table 1 and online supplemental table S7). Furthermore, 1102 variants were identified with a moderate evidence level of PM2, comprising 1002 disease-causing variants and 100 non-pathogenic variants. The accuracy was 0.531, PPV was 0.910, sensitivity was 0.303, specificity was 0.945, and LR^+ was 5.531 (95% CI 4.575 to 6.818), reaching the supporting evidence level. In addition, using the OP thresholds based on an empirically prior of 10% ($O_{PSu} \geq 2.08$, $O_{PM} \geq 4.33$, $O_{PS} \geq 18.7$, $O_{PVSt} \geq 350$, $O_{BSu} \leq 0.48$, $O_{BM} \leq 0.23$, $O_{BS} \leq 0.053$ and $O_{BVSt} \leq 0.0029$), PM2 could reach moderate evidence for pathogenicity.

Calibration of in-silico tools for predicting pathogenicity for the *PTEN* gene

We systematically evaluated 38 prediction scores included in the dbNSFP database to establish calibrated PP3/BP4 thresholds for interpreting *PTEN* missense variants. The distribution of prediction scores between non-pathogenic and disease-causing variants in the *PTEN* dataset across all 38 tools was shown in online supplemental figure S4. Twelve tools (SiPhy, phastCons, phyloP, GERP++, fitCons, GenoCanyon, fathmm_MKL,

Table 1 Odds of pathogenicity and corresponding posterior probability values for each evidence strength across the three datasets included in this study

Evidence strength	ClinGen Curated Variants dataset		ClinVar 2019 dataset		PTEN dataset	
	OP	Post_P of pathogenicity or benignity	OP	Post_P of pathogenicity or benignity	OP	Post_P of pathogenicity or benignity
PVS	2490	0.985	1132	0.981	1042	0.981
PS	49.90	0.560	33.65	0.608	32.28	0.613
PM	7.064	0.153	5.800	0.211	5.682	0.218
PP	2.658	0.064	2.408	0.100	2.384	0.105
BP	0.376	0.990	0.415	0.981	0.420	0.980
BM	0.142	0.996	0.172	0.992	0.176	0.991
BS	0.020	0.999	0.030	0.999	0.031	0.998
BVS	0.0004	1.000	0.001	1.000	0.001	1.000

BM, moderate for benignity; BP, supporting for benignity; BS, strong for benignity; BVS, very strong for benignity; OP, odds of pathogenicity; PM, moderate for pathogenicity; Post_P, posterior probability; PP, supporting for pathogenicity; PS, strong for pathogenicity; PVS, very strong for pathogenicity.

DANN, LIST_S2, fathmm_XF, MutationTaster and LRT) were excluded from downstream analysis due to over-mixed prediction scores for both non-pathogenic and disease-causing variants. The remaining 26 tools showed an area under the curve (AUC) values ranging from 0.782 to 0.946 (online supplemental figure S5), with the Evolutionary model of Variant Effect (EVE; AUC=0.946) exhibiting the best performance, followed closely by Rare Exome Variant Ensemble Learner (REVEL; AUC=0.945) and Variant Effect Scoring Tool (VEST; AUC=0.943).

The local posterior probability-based approach proposed by ClinGen SVI allowed for identifying thresholds corresponding to different strengths of evidence. This approach was applied to calibrate the top 10 performing tools, including AlphaMissense,²⁷ BayesDel,²⁸ ClinPred,²⁹ EVE,³⁰ M-CAP,³¹ MetaLR,³² MetaRNN,³³ MVP,³⁴ REVEL³⁵ and SIFT.³⁶ We identified thresholds for supporting and moderate evidence of pathogenicity (PP3) and benignity (BP4) for all tools, except for MetaLR and MVP, which did not reach moderate evidence for BP4. Interestingly, the local posterior probability curves demonstrated that, at appropriate thresholds, several tools could provide strong evidence for pathogenicity (ClinPred, M-CAP, MetaLR, MetaRNN) or benignity (AlphaMissense), as illustrated in figure 4, online supplemental figure S6 and table 2. When applying a commonly used prior probability of 10%, we observed that ClinPred, M-CAP, MetaLR, MetaRNN, MVP and REVEL achieved strong evidence for pathogenicity, while AlphaMissense, ClinPred and EVE could reach strong evidence for benignity (online supplemental figure S7 and table S8).

DISCUSSION

The ClinGen Bayesian framework offers a solid mathematical basis for refining the ACMG/AMP criteria within clinical laboratory practice. However, to our knowledge, there is a lack of freely available software tools that provide similar functionalities for the academic community. In this article, we present *BayesQuantify*, an R package designed to enhance the flexibility and convenience of quantifying the strength of evidence and establishing corresponding thresholds for the ACMG/AMP criteria based on the Bayesian framework. Evaluation across three independent datasets confirms the utility of *BayesQuantify* as a user-friendly tool for improving the specification and stratification of the ACMG/AMP criteria.

BayesQuantify offers an objective assessment of whether various forms of evidence align with the relative weights specified by the Bayesian framework. Our analysis indicated that the

PM2 criterion could reach ‘supporting,’ or ‘moderate,’ evidence, varying by prior probability. However, the ClinGen SVI Working Group recommended that the PM2 criterion should be downgraded from moderate strength level to supporting strength level (<https://clinicalgenome.org/docs/pm2-recommendation-for-absence-rarity/>). To reduce the potential impact of PM2 downgrading in variant classification, the ClinGen SVI also proposed novel evidence combination rules beyond the 2015 ACMG/AMP guidelines, combining a Very Strong (e.g., PVS1) and a Supporting criterion (e.g., PM2_Supporting) to achieve a LP classification (Bayesian posterior probability=0.988). As the ClinGen Curated Variants dataset comprises variants highly selected by expert panels, we recommend that further systematic evaluation and calibration of the evidence weight for PM2 is necessary.

The Bayesian framework implemented in *BayesQuantify* extends the quantification of evidence strength from categorical to continuous variables. We replicated Pejaver *et al*’s findings, demonstrating that BayesDel, MutPred2, REVEL and VEST4 achieve strong evidence thresholds for pathogenicity (PP3) and moderate evidence for benignity (BP4) in missense variant classification. The slight discrepancies between our results and those of Pejaver *et al* can be attributed to variations in the bootstrap process. Although these findings are applicable across various genes, a genome-wide approach may mask the heterogeneous performance of predictors within distinct genes.^{37 38} As such, gene-specific calibrations are crucial for improving accuracy and precision in variant classification.

Our study attempted to calibrate the PP3 and BP4 criteria for missense variants in the *PTEN* gene. Germline pathogenic variants in *PTEN* are associated with a highly heterogeneous clinical spectrum and are observed in individuals with conditions such as Cowden syndrome, Bannayan-Riley-Ruvalcaba syndrome and other phenotypes, including macrocephaly and autism, collectively referred to as *PTEN* hamartoma tumour syndrome.^{39 40} In 2018, the ClinGen *PTEN* Expert Panel published gene-specific criteria for *PTEN* variant curation, incorporating disease-specific modifications and adjustments to the strength of evidence across 19 criteria.⁴¹ However, the guidelines did not recommend specific thresholds for applying the PP3/BP4 criteria to *PTEN* missense variants, creating a critical gap in variant interpretation. According to our results, we recommend using MetaRNN or AlphaMissense to determine evidence for codes PP3 and BP4 criteria, as these tools provide strong or moderate evidence for pathogenicity and benignity.^{27 33} Both tools showed high AUC

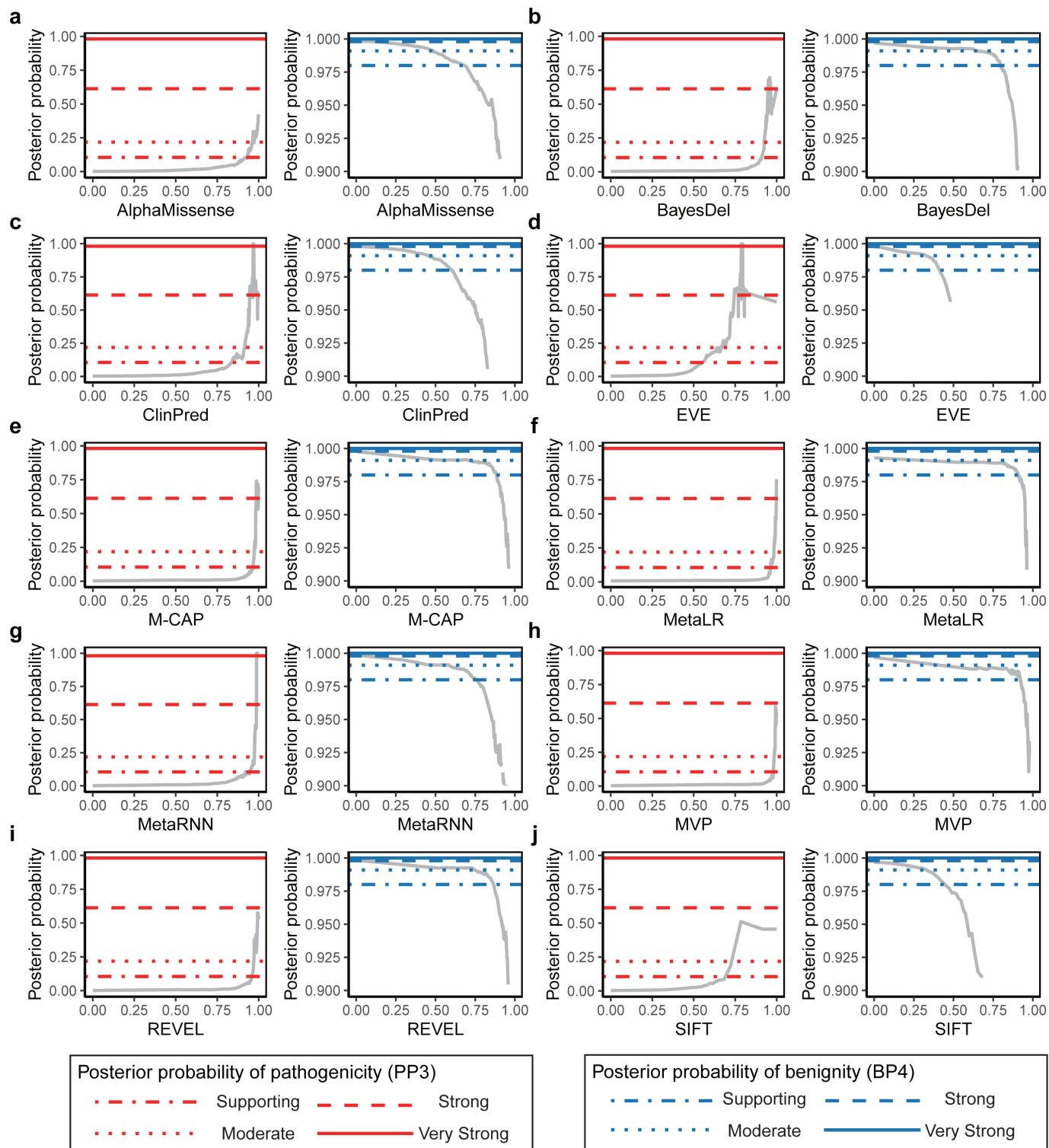


Figure 4 : Local posterior probability curves for ten in-silico tools in the PTEN dataset. The horizontal lines depict the posterior probability thresholds for supporting, moderate, strong, and very strong evidence (red for pathogenicity and blue for benignity). The grey curves represent one-sided 95% confidence intervals calculated from 10 000 bootstrap samples of the PTEN dataset. SIFT, sorting intolerant from tolerant; M-CAP, mendelian clinically applicable pathogenicity; MVP, missense variant pathogenicity prediction; EVE, evolutionary model of variant effect; REVEL, rare exome variant ensemble learner.

values (above 0.93), optimising the strength of evidence while minimising false positives in the supporting and moderate categories. Given the absence of an independent validation dataset, future studies are needed to confirm these thresholds.

A critical consideration when applying *BayesQuantify* is the selection of prior probability. As described in the Methods section, the approaches used to calculate prior probability involve specific considerations and trade-offs. The optimal strategy involves calibration using disease-specific or gene-specific

Table 2 Estimated threshold ranges corresponding to the four pathogenic and four benign intervals for the *PTEN* gene

Method	Benign (BP4)				Pathogenic (PP3)			
	Very strong	Strong	Moderate	Supporting	Supporting	Moderate	Strong	Very strong
AlphaMissense	–	≤0.060	(0.060, 0.453]	(0.453, 0.643]	[0.935, 0.974)	≥0.974	–	–
BayesDel	–	–	≤0.638	(0.638, 0.794]	[0.908, 0.925)	≥0.925	–	–
ClinPred	–	–	≤0.447	(0.447, 0.564]	[0.836, 0.917)	[0.917, 0.999)	≥0.999	–
EVE	–	–	≤0.324	(0.324, 0.385]	[0.567, 0.668)	≥0.668	–	–
M-CAP	–	–	≤0.686	(0.686, 0.883]	[0.967, 0.973)	[0.973, 0.998)	≥0.998	–
MetaLR	–	–	–	≤0.907	[0.965, 0.983)	[0.983, 0.999)	≥0.999	–
MetaRNN	–	–	≤0.581	(0.581, 0.745]	[0.949, 0.976)	[0.976, 0.986)	[0.986, 0.987)	≥0.987
MVP	–	–	–	≤0.912	[0.980, 0.985)	≥0.985	–	–
REVEL	–	–	≤0.744	(0.744, 0.865]	[0.959, 0.972)	≥0.972	–	–
SIFT	–	–	≤0.328	(0.328, 0.440]	[0.722, 0.785)	≥0.785	–	–

EVE, evolutionary model of variant effect; M-CAP, mendelian clinically applicable pathogenicity; MVP, missense variant pathogenicity prediction; REVEL, rare exome variant ensemble learner; SIFT, sorting intolerant from tolerant.

cohorts that comprehensively capture variant spectra across clinically relevant genes.^{17,23} However, population-specific allele frequencies introduce a significant limitation: calibrated thresholds derived from one population may lack generalisability to others. We therefore emphasise that establishing population-specific cohorts is essential for precise evidence quantification. In clinical practice, well-curated cohorts for Mendelian disorders remain scarce. Where such resources are unavailable, prior probability estimation becomes context-dependent. For datasets lacking a predefined catalogue (e.g., functional assay data), a commonly used empirical prior of 10% is recommended. Additionally, when datasets contain fewer than 25 pathogenic or benign variants, the limited number of variants may increase the variance around the calibration thresholds.²³ In such cases, we recommend that calibration be performed subsequently when sufficient data has been accumulated. Given the frequent scarcity of pathogenic and benign variants for individual genes, aggregating variants across genes associated with the same function, pathway or phenotype, and calibrating at this aggregated level, is an effective strategy to improve calibration accuracy, despite the limited availability of gene-level data. Developing alternative methods for estimating prior probability of pathogenicity across diverse contexts represents a valuable direction for future research.

A limitation worth discussing is the assumption within the Bayesian framework that each categorical type of evidence at the same evidence strength level had the same mathematical support. Given the complexity of interpreting genetic evidence, this simplifying assumption is reasonable but may not fully capture the nuances involved. Previous studies have identified dependencies and correlations between evidence rules, such as those observed in BS3/BP4 and PVS1/PP3.^{42,43} To decrease associated double-counting risks, *BayesQuantify* incorporates the *evaluation_variant()* function, which automatically identifies variants exhibiting high correlated ACMG/AMP evidence. This enables users to prioritise variants for re-evaluation and reduce redundancy. Further refinement could be achieved by estimating evidence-specific weights using real-world datasets with confirmed true positives/negatives variants, as implemented for breast and colorectal cancer genes.^{16,44} Finally, given the potential misclassification in public databases,^{45–47} we stress that user-curated datasets remain essential for optimal evidence calibration when applying *BayesQuantify*.

In summary, this study introduces *BayesQuantify*, a novel, open-access tool that facilitates a shift toward a more quantitative

and objective approach to ACMG/AMP evidence refinement, moving away from predominantly subjective assessments. By calibrating and expanding evidence strength, *BayesQuantify* could improve the accuracy and consistency of genetic variant classification, thereby supporting the resolution of VUS.

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REFERENCES

- Giles HH, Hegde MR, Lyon E, et al. The Science and Art of Clinical Genetic Variant Classification and Its Impact on Test Accuracy. *Annu Rev Genomics Hum Genet* 2021;22:285–307.
- Marian AJ. Clinical Interpretation and Management of Genetic Variants. *JACC Basic Transl Sci* 2020;5:1029–42.

- 3 Pepin MG, Murray ML, Bailey S, *et al.* The challenge of comprehensive and consistent sequence variant interpretation between clinical laboratories. *Genet Med* 2016;18:20–4.
- 4 Richards S, Aziz N, Bale S, *et al.* Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genet Med* 2015;17:405–24.
- 5 Niehaus A, Azzariti DR, Harrison SM, *et al.* A survey assessing adoption of the ACMG-AMP guidelines for interpreting sequence variants and identification of areas for continued improvement. *Genet Med* 2019;21:1699–701.
- 6 Amendola LM, Jarvik GP, Leo MC, *et al.* Performance of ACMG-AMP Variant-Interpretation Guidelines among Nine Laboratories in the Clinical Sequencing Exploratory Research Consortium. *Am J Hum Genet* 2016;98:1067–76.
- 7 Patel MJ, DiStefano MT, Oza AM, *et al.* Disease-specific ACMG/AMP guidelines improve sequence variant interpretation for hearing loss. *Genet Med* 2021;23:2208–12.
- 8 Amendola LM, Muenzen K, Biesecker LG, *et al.* Variant Classification Concordance using the ACMG-AMP Variant Interpretation Guidelines across Nine Genomic Implementation Research Studies. *Am J Hum Genet* 2020;107:932–41.
- 9 Tavtigian SV, Greenblatt MS, Harrison SM, *et al.* Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework. *Genet Med* 2018;20:1054–60.
- 10 Pejaver V, Byrne AB, Feng B-J, *et al.* Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria. *Am J Hum Genet* 2022;109:2163–77.
- 11 Villani RM, McKenzie ME, Davidson AL, *et al.* Regional-specific calibration enables application of computational evidence for clinical classification of 5' cis-regulatory variants in Mendelian disease. *Am J Hum Genet* 2024;111:1301–15.
- 12 Brünger T, Ivaniuk A, Pérez-Palma E, *et al.* Pathogenic paralogous variants can be used to apply the acmg ps1 and pm5 variant interpretation criteria. *medRxiv* [Preprint].
- 13 Walker LC, Hoya M de la, Wiggins GAR, *et al.* Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI Splicing Subgroup. *Am J Hum Genet* 2023;110:1046–67.
- 14 Brnich SE, Abou Tayoun AN, Couch FJ, *et al.* Recommendations for application of the functional evidence P53/BS3 criterion using the ACMG/AMP sequence variant interpretation framework. *Genome Med* 2019;12:3.
- 15 Wilcox EH, Sarmady M, Wulf B, *et al.* Evaluating the impact of in silico predictors on clinical variant classification. *Genet Med* 2022;24:924–30.
- 16 Clark KA, Paquette A, Tao K, *et al.* Comprehensive evaluation and efficient classification of BRCA1 RING domain missense substitutions. *Am J Hum Genet* 2022;109:1153–74.
- 17 Liu S, Zhong M, Huang Y, *et al.* Quantitative thresholds for variant enrichment in 13,845 cases: improving pathogenicity classification in genetic hearing loss. *Genome Med* 2023;15:116.
- 18 Rehm HL, Berg JS, Brooks LD, *et al.* ClinGen—the Clinical Genome Resource. *N Engl J Med* 2015;372:2235–42.
- 19 Marill KA, Chang Y, Wong KF, *et al.* Estimating negative likelihood ratio confidence when test sensitivity is 100%: A bootstrapping approach. *Stat Methods Med Res* 2017;26:1936–48.
- 20 Karczewski KJ, Francioli LC, Tiao G, *et al.* The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature New Biol* 2020;581:434–43.
- 21 Landrum MJ, Lee JM, Benson M, *et al.* ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Res* 2018;46:D1062–7.
- 22 Stenson PD, Mort M, Ball EV, *et al.* The Human Gene Mutation Database (HGMD®): optimizing its use in a clinical diagnostic or research setting. *Hum Genet* 2020;139:1197–207.
- 23 Bhat V, Yu T, Brown L, *et al.* Extracting and calibrating evidence of variant pathogenicity from population biobank data. *Am J Hum Genet* 2025;112:1805–17.
- 24 Liu X, Li C, Mou C, *et al.* dbNSFP v4: a comprehensive database of transcript-specific functional predictions and annotations for human nonsynonymous and splice-site SNVs. *Genome Med* 2020;12:103.
- 25 McLaren W, Gil L, Hunt SE, *et al.* The Ensembl Variant Effect Predictor. *Genome Biol* 2016;17:122.
- 26 Robin X, Turck N, Hainard A, *et al.* pROC: an open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinformatics* 2011;12:77.
- 27 Cheng J, Novati G, Pan J, *et al.* Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science* 2023;381:eadg7492.
- 28 Feng BJ. PERCH: A Unified Framework for Disease Gene Prioritization. *Hum Mutat* 2017;38:243–51.
- 29 Alirezai N, Kernohan KD, Hartley T, *et al.* ClinPred: Prediction Tool to Identify Disease-Relevant Nonsynonymous Single-Nucleotide Variants. *Am J Hum Genet* 2018;103:474–83.
- 30 Frazer J, Notin P, Dias M, *et al.* Disease variant prediction with deep generative models of evolutionary data. *Nature New Biol* 2021;599:91–5.
- 31 Jagadeesh KA, Wenger AM, Berger MJ, *et al.* M-CAP eliminates a majority of variants of uncertain significance in clinical exomes at high sensitivity. *Nat Genet* 2016;48:1581–6.
- 32 Dong C, Wei P, Jian X, *et al.* Comparison and integration of deleteriousness prediction methods for nonsynonymous SNVs in whole exome sequencing studies. *Hum Mol Genet* 2015;24:2125–37.
- 33 Li C, Zhi D, Wang K, *et al.* MetaRNN: differentiating rare pathogenic and rare benign missense SNVs and InDels using deep learning. *Genome Med* 2022;14:115.
- 34 Qi H, Zhang H, Zhao Y, *et al.* MVP predicts the pathogenicity of missense variants by deep learning. *Nat Commun* 2021;12:510.
- 35 Ioannidis NM, Rothstein JH, Pejaver V, *et al.* REVEL: An Ensemble Method for Predicting the Pathogenicity of Rare Missense Variants. *Am J Hum Genet* 2016;99:877–85.
- 36 Ng PC, Henikoff S. SIFT: Predicting amino acid changes that affect protein function. *Nucleic Acids Res* 2003;31:3812–4.
- 37 Brock DC, Wang M, Hussain HMJ, *et al.* Comparative analysis of in-silico tools in identifying pathogenic variants in dominant inherited retinal diseases. *Hum Mol Genet* 2024;33:945–57.
- 38 Tejura M, Fayer S, McEwen AE, *et al.* Calibration of variant effect predictors on genome-wide data masks heterogeneous performance across genes. *Am J Hum Genet* 2024;111:2031–43.
- 39 Mester J, Eng C. When overgrowth bumps into cancer: the PTEN-opathies. *Am J Med Genet C Semin Med Genet* 2013;163C:114–21.
- 40 Yehia L, Keel E, Eng C. The Clinical Spectrum of PTEN Mutations. *Annu Rev Med* 2020;71:103–16.
- 41 Mester JL, Ghosh R, Pesaran T, *et al.* Gene-specific criteria for PTEN variant curation: Recommendations from the ClinGen PTEN Expert Panel. *Hum Mutat* 2018;39:1581–92.
- 42 Kanavy DM, McNulty SM, Jairath MK, *et al.* Comparative analysis of functional assay evidence use by ClinGen Variant Curation Expert Panels. *Genome Med* 2019;11:77.
- 43 Garrett A, Durkie M, Callaway A, *et al.* Combining evidence for and against pathogenicity for variants in cancer susceptibility genes: CanVIG-UK consensus recommendations. *J Med Genet* 2021;58:297–304.
- 44 Easton DF, Deffenbaugh AM, Pruss D, *et al.* A systematic genetic assessment of 1,433 sequence variants of unknown clinical significance in the BRCA1 and BRCA2 breast cancer-predisposition genes. *Am J Hum Genet* 2007;81:873–83.
- 45 Wright CF, Eberhardt RY, Constantinou P, *et al.* Evaluating variants classified as pathogenic in ClinVar in the DDD Study. *Genet Med* 2021;23:571–5.
- 46 Sharo AG, Zou Y, Adhikari AN, *et al.* ClinVar and HGMD genomic variant classification accuracy has improved over time, as measured by implied disease burden. *Genome Med* 2023;15:51.
- 47 Shah N, Hou Y-C, Yu H-C, *et al.* Identification of Misclassified ClinVar Variants via Disease Population Prevalence. *Am J Hum Genet* 2018;102:609–19.

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